

Tomas Palacios, who heads Centre for Graphene Devices and 2D systems at Massachusetts Institute of Technology (MIT), is one such scientist exploring the applications of graphene. He has several radical ideas, of which one is described by him as display on demand.

He believes that graphene could turn everyday objects like coffee cups and shoes into devices that can easily compile and transmit information. "Basically, everything around us will be able to convert itself into a display on demand," he said during a newspaper interview.

To implement this, his team is developing a 3D printer that uses graphene as its ink. These objects would

wearables and other next-generation electronic devices. There are rumours that the company is already testing graphene based touchscreens.

Bodle's bid for power

If flexibility is one requirement for future displays, then power efficiency is equally important. If you are bound to have devices all around you, you would certainly not want to keep charging all of these.

Bodle Technologies, a UK based startup, is working on a technology that could be used to make super-efficient displays that require almost no power at all. Their tech revolves around ultra-thin structures made using well-known and well-characterised materials including phase-change materials. This structure can manipulate light using electrical, optical or mechanical means.

According to the company, "It can be used to filter, steer or dim light at a flick of a switch and, most importantly, consumes very little or no power after switching." This optoelectronic framework based platform technology can be used for developing low-power, flexible displays, smartwindows and much more.

Personal home theatre, anyone

Head-mounted displays, which were clunky and cumbersome at one point of time, are now sleek, lightweight and power-efficient. No wonder these devices, like Samsung GearVR, are touted to be more than gaming devices. These virtual reality (VR) displays have the potential to turn into personal home theatres, giving you an immersive, comfortable and personalised movie experience.

However, holographic displays that are likely to become popular in the next five to ten years are likely to put today's 3D, VR and AR displays to shame. Holographic and volumetric displays appear to give an object mass. So you actually feel as if you

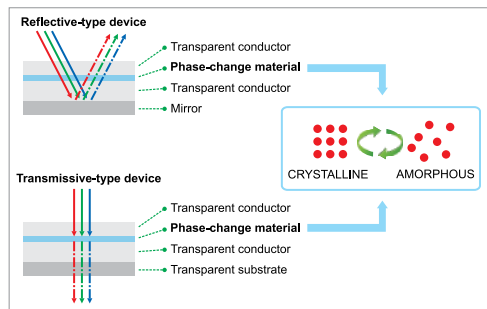


Fig. 2: Basic working principle of Bodle Technologies' optoelectronic framework based platform technology

have electrical intelligence built into these. Graphene with its electrical and optical properties could enable the object itself to turn into a display when required.

There have been earlier attempts to make awesome displays with graphene. Researchers from University of Manchester and University of Sheffield, for example, came up with a prototype semi-transparent, flexible, graphene based LED display that could be used in phones, tablets and even televisions. The super-thin display was just 10 atoms to 40 atoms thick and could emit a sheet of light across its entire surface.

Samsung, which has developed a method of synthesising large-area, single-crystal wafer-scale graphene, also believes that graphene is the perfect material for use in flexible displays,

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